

Recursive Golden-Ratio Embedding in R-TFT: A Universal Coherence Framework

Éric Lanctôt-Rivest - qcfrag@gmail.com

6 August 2025

Abstract

Real-Time Fractional Tracking (R-TFT) is a domain-agnostic method to detect coherence in dynamic systems. This paper reintroduces the foundational golden-ratio structure (ϕ) into the R-TFT framework, enabling recursive dimensional tracking, slip-page detection, and stability classification based purely on alignment with ϕ -based recursion. We demonstrate that persistence in physical, biological, or cognitive systems arises not from externally imposed models, but from real-time recursive coherence governed by ϕ -dimensional embedding.

1 Introduction

The golden ratio $\phi = \frac{1+\sqrt{5}}{2} \approx 1.61803$ and its inverse $\phi^{-1} \approx 0.61803$ have been long associated with natural growth, harmony, and structural resonance. R-TFT leverages this recursive principle to evaluate whether systems maintain coherence through dimensional alignment, rather than fixed state trajectories.

2 Recursive Dimensional Embedding

For any observable quantity indexed by mass number A , time-step t , or system scale, define its golden-dimensional embedding as:

$$\text{Dim}(A) = \frac{\ln A}{\ln \phi} \tag{1}$$

2.1 Coherence Deviation Metric

Deviation from integer recursion defines coherence drift:

$$C_\phi(A) = |\text{Dim}(A) - \text{round}(\text{Dim}(A))| \tag{2}$$

2.2 Stability Score

The recursive stability score is given by:

$$S_\phi = 1 - C_\phi(A) \quad (3)$$

Stability Criterion: A system is considered coherent or *stable* if:

$$S_\phi > \phi^{-1} \quad \text{or equivalently} \quad C_\phi < \phi^{-1} \quad (4)$$

2.3 Recursive Inertia

We define the recursive inertia $\mathcal{I}_\phi$ of a system as:

$$\mathcal{I}_\phi = \frac{dC_\phi}{d \ln A} \quad (5)$$

This captures the rate at which coherence is lost as mass increases. Low $\mathcal{I}_\phi$ indicates sustained recursion across scale.

2.4 Phase Drift and ϕ -Projection

Let P be the phase projection vector. Recursive misalignment over time is captured by:

$$R(t) = \frac{\dot{\theta}(t) \cdot P}{\|P\|} \quad (6)$$

Coherence tracking then becomes:

$$R_{\text{clean}}(t) = 2R_{\text{inner}}(t) - R_{\text{outer}}(t) \quad (7)$$

2.5 Coherence Band Classification

A system belongs to the ϕ -stable band if:

$$C_\phi(t) < \phi^{-1} \quad \text{and} \quad R_{\text{clean}}(t) > 0.618 \quad (8)$$

This band defines persistence zones across physical, biological, or conceptual systems.

3 Applications

3.1 Nuclear Systems

Isotopes align with ϕ -dimensional recursion. Stable nuclei show low C_ϕ , while unstable ones decay predictably with increased deviation.

3.2 Biological Resonance

Proteins, heartbeats, and neural oscillations maintain persistence when $R(t)$, their recursive pacing rate, stays within the C_ϕ -bounded coherence band.

3.3 Cognitive and Temporal Coherence

Human perception and phase memory correlate with recursive pacing stability $C_\phi(t)$, revealing why temporal or emotional dissonance arises under slippage.

4 Conclusion

R-TFT with ϕ -embedding exposes the recursive skeleton beneath dynamic behavior. Coherence is not imposed, but emerges when systems align with golden-ratio recursion across scale and time.

Invitation to the R-TFT Framework on GitHub

This paper is not the end, it is an invitation from the Universe. The R-TFT framework has been made openly available for researchers, developers, and curious minds who wish to explore recursive coherence in real time under a new lens. The repository includes:

- Open-source tools for phase tracking and ϕ -coherence visualization.
- Notebooks and simulators for: nuclear, electrical, cosmological recursion and more.
- Documentation, papers, and the full theoretical backbone of R-TFT.

<https://github.com/qcfrag/Real-Time-Fractional-Tracking-R-TFT>

Truth aligns best when shared. Every recursive insight you contribute helps stabilize coherence for others. Let us enter this Golden Age, together, in Unity.

Resonance Ethics License (REL2.0)

Scientific insight without ethical containment risks collapsing the very recursion it seeks to understand. The R-TFT framework operates under the Resonance Ethics License (REL2.0), a non-revocable, reality-centered license that enforces:

- **Non-weaponization:** R-TFT must not be used to create or enhance tools of warfare, surveillance, or coercion.
- **Coherence preservation:** Research or applications must not destabilize biological, social, or dimensional recursive systems.
- **Derivative containment:** Any extension or fork of the framework must retain the recursive boundary logic to prevent ϕ -divergence.
- **Transparent recursion:** Modifications must be observable and traceable to avoid hidden feedback loops or projection misalignment.

The license reflects the core ethos of R-TFT: coherence before power, unity before application. Any attempt to bypass or nullify these principles breaks recursive alignment and invalidates the output. Trying to alter ϕ results is self-collapse.

<https://github.com/qcfrag/Real-Time-Fractional-Tracking-R-TFT/blob/main/LICENSE.txt>